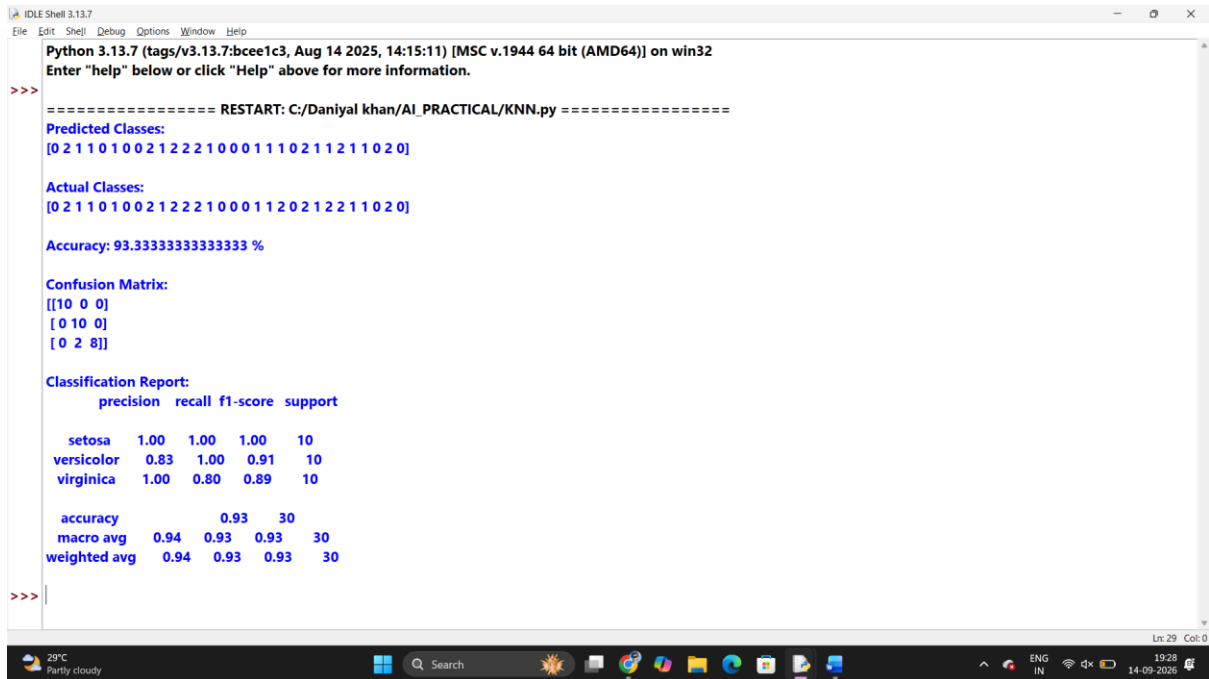


SHETH LUJ COLLEGE

K-Nearest Neighbors (K-NN)

- ❑ Implement the K-NN algorithm for classification or regression.
- ❑ Apply the K-NN algorithm to a given dataset and predict the class or value for test data.
- ❑ Evaluate the accuracy or error of the predictions and analyze the results.



```
Python 3.13.7 (tags/v3.13.7:bcee1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Daniyal khan/AI_PRACTICAL/KNN.py =====
Predicted Classes:
[0 2 1 1 0 1 0 0 2 1 2 2 1 0 0 0 1 1 1 0 2 1 1 2 1 1 0 2 0]

Actual Classes:
[0 2 1 1 0 1 0 0 2 1 2 2 1 0 0 0 1 1 2 0 2 1 2 2 1 1 0 2 0]

Accuracy: 93.33333333333333 %

Confusion Matrix:
[[10 0 0]
 [0 10 0]
 [0 2 8]]

Classification Report:
      precision    recall  f1-score   support

 setosa      1.00      1.00      1.00        10
 versicolor  0.83      1.00      0.91        10
 virginica    1.00      0.80      0.89        10

 accuracy          0.93        30
 macro avg      0.94      0.93      0.93        30
 weighted avg    0.94      0.93      0.93        30

>>>
```